

PRANAV SINGH

Full Stack Developer | React.js | Next.js | Node.js | TypeScript

pranasinghind@gmail.com | 8077004381 | linkedin.com/in/pranav-signh | github.com/PRANAV-SINGH-CSE

TECHNICAL SKILLS

Languages: Java, C, C++, JavaScript, TypeScript, Python, SQL

Frontend: HTML5, CSS3, React.js, Next.js, Tailwind CSS

Backend: Node.js, Flask, FastAPI, REST APIs, WebSockets

Databases: MySQL, Firebase, Firestore, Realtime Database

AI/ML: OpenAI API, Anthropic API, Gemini API, LLM Integration, Prompt Engineering

Tools: Git, GitHub, GitHub Actions, CI/CD, Vercel

PROJECTS

Sprinto — Real-Time Collaborative Workspace & Kanban Platform

Jul 2026 – Aug 2026

Tech: Next.js, TypeScript, Firebase, Tailwind CSS, @dnd-kit, IndexedDB

Live Demo: <https://sprinto-move.vercel.app/> | **GitHub:** <https://github.com/PRANAV-SINGH-CSE/Sprinto>

- Developed a real-time collaborative Kanban platform using Next.js, TypeScript, and Firebase, enabling simultaneous multi-user board editing with live synchronization.
- Optimized real-time synchronization to achieve <1ms local UI updates and ~4.5ms median cross-client propagation, decoupled from the ~350ms average cloud write commit.
- Cut repeat-visit board load time from ~400ms (Firestore query) to ~5ms using an IndexedDB local cache layer.

LiveVoice AI — Real-Time Speech Intelligence & Desktop HUD Platform

Jun 2026 – Jul 2026

Tech: Python, FastAPI, WebSockets, AsyncIO, PyQt5, Win32 API, JavaScript (ES6+), HTML5, CSS3, OpenAI API, Anthropic API, Faster-Whisper

- Engineered a real-time voice assistant that transcribes speech and streams AI responses live, achieving sub-700ms latency from end of speech to first spoken token and a sub-1-second full response loop, measured via P50/P90 profiling across the audio-to-LLM pipeline.
- Tuned the speech-to-text stage to run at a 0.28x real-time factor (3.5x faster than speech) with Faster-Whisper, while the LLM streamed responses at ~197 tokens/sec.
- Designed a transparent, always-on-top desktop overlay controlled by global hotkeys for use during screen shares and calls.
- Implemented an async FastAPI backend that extracts key questions from streamed transcripts in real time, with full session history exportable to Markdown/JSON.

ZentiqAI — AI Chat Web Application

Feb 2026 – Apr 2026

Tech: Python (Flask), JavaScript, Firebase, HTML, CSS

Live Demo: <https://zentiqai.vercel.app/> | **GitHub:** https://github.com/PRANAV-SINGH-CSE/ZentiqAI_Chat_Bot

- Created a full stack AI chat app with persistent, multi-session conversations and image upload, letting users pick up a conversation from any device.
- Engineered concurrent session handling on the backend so multiple users can chat at the same time without their session data colliding.
- Restructured the frontend into a modular, class-based JavaScript architecture for easier feature extension.

EDUCATION

Lovely Professional University (LPU)

B.Tech, Computer Science and Engineering; TGPA:8.4